

Revised Light Project Spec:

REVISED/REMOVED PIECES:

- Removed all variables including costs and weather, the app will not create recommendations based on predicted weather data, and users will not be able to filter by cost.
- There will not be a population heatmap on the map to overlay projected event density.
- There will not be local storage where users can store favorites in an account info, saving events for a later reference.
- Events will come from Ticketmaster and Eventbrite.
- There will not be a generated recommendations list for the user based on favorite activities.

Modified Spec:

General description of the project

This project is an interactive Kansas City event discovery application designed to help residents, students, and tourists find things to do in one place. Rather than searching multiple websites, users can browse concerts, festivals, museums, restaurants, coffee shops, parks, and other local activities through a single platform. The goal is to simplify event discovery and make it easier for people to explore the Kansas City area based on their interests and location.

The application will collect data from the Ticketmaster Discovery API, Eventbrite. Information such as event names, dates, locations, and categories will be processed and combined into a searchable database. Users will be able to filter activities by date, time, or category while the system removes duplicate events and organizes information into a user-friendly format.

POSSIBLE: A key feature of the project is an interactive map built with Leaflet and OpenStreetMap. Instead of only displaying individual event pins, the map will help users explore Kansas City neighborhoods such as Crossroads, River Market, Westport, and the Plaza. Events, attractions, and activity hotspots will be visualized through map markers and neighborhood overlays allowing users to quickly identify areas that match their interests and discover new experiences throughout the city.

Task Vignettes (User activity "flow")

Vignette 1: Discovering Events Through the Interactive Map

A college student is looking for something to do on a Friday night but does not know where to start. They open the Kansas City Event Explorer and are presented with an interactive map showing neighborhoods such as Crossroads, River Market, Westport, and the Plaza. The student selects the "Live Music" category and immediately sees upcoming concerts, local performances, and music festivals displayed on the map. As they zoom in, event clusters separate into individual markers, making it easier to identify activities near their preferred area.

The student clicks on an event marker in the Crossroads district and views additional details, including the event description, date, time, location, and ticket information.

Technical Details

- Retrieve event data from Ticketmaster and Eventbrite APIs.
- Store event names, categories, locations, dates, and ticket links.
- Use Leaflet.js and OpenStreetMap to display an interactive map.
- Convert event locations into latitude and longitude coordinates.
- Cluster nearby markers to reduce visual clutter.

Vignette 2: Exploring a Neighborhood

A tourist visiting Kansas City for the weekend wants to find a neighborhood with multiple activities within walking distance. Instead of searching event-by-event, they select the River Market neighborhood on the map. The application highlights upcoming events occurring within that district.

Technical Details

- Import neighborhood boundary data using GeoJSON files.
- Display neighborhood overlays on the map.
- Associate events and attractions with neighborhood polygons.

Vignette 3: Finding Personalized Recommendations

A local resident wants affordable activities for the weekend. Upon opening the application, they choose preferences such as "Free Activities," "Outdoor Events," and a maximum budget of \$20. The system filters hundreds of available events and displays only those that match the selected criteria. A recommendation list appears alongside the map, highlighting activities that align with the user's interests.

The resident discovers a free outdoor concert and a local art walk occurring in nearby neighborhoods. After reviewing the recommendations, they save both activities and share one with a friend. The application helps the user narrow down choices and discover events they may not have found through traditional searches.

Technical Details

- Apply filtering logic based on category, date, and distance.
- Store user preferences during the session.
- Update map markers dynamically as filters change.
- Remove duplicate events collected from multiple APIs.